

Calibrated Analog Joystick

Preserve raw evidence, shape two axes, debounce one selection

Three observations; one deterministic snapshot

Lesson	031 — component	Time	150–210 minutes
Prerequisites	Lessons 002, 007, and 008	Board	Arduino Mega 2560
Evidence	TP-X, TP-Y, RGB preview	Status	host verified; bench open
Energy class	E1: USB, passive joystick, resistor-limited RGB LED		

Specimen gate. The official Elegoo manifests authorize a “Joystick module” family, not a universal pin order or catalog identity. Before wiring, read the labels on the exact unpowered board and establish its supply rating and signal roles. Do not infer KY-023, pin order, polarity, or axis direction from board color or an internet photograph. Unknown markings stop the physical procedure; they do not block the host lesson.

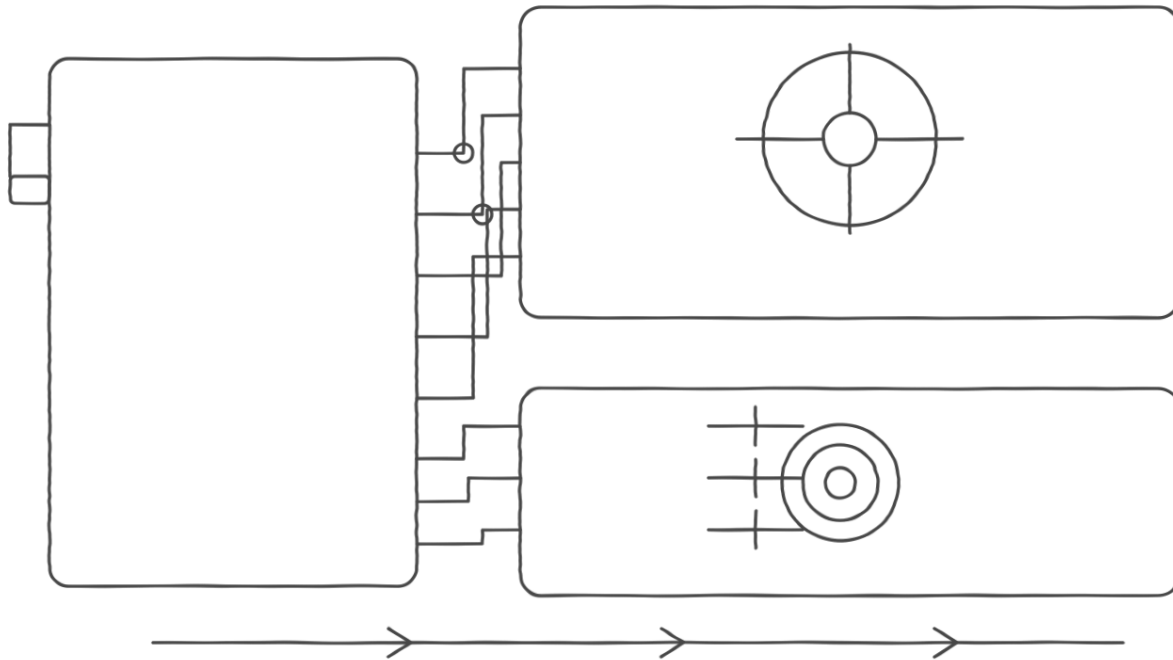
1 Mission and learning outcomes

X raw + Y raw + switch raw → calibrated snapshot → RGB direction and magnitude.

By the end, you can:

- distinguish raw ADC evidence from normalized position;
- map asymmetric travel around a measured center without floating point;
- explain why a center dead zone is not discarded evidence;
- debounce selection with caller-supplied time;
- predict initialization rollback and reverse-order shutdown;
- prove acquisition separately from input-mode safe state.

2 Orientation and unpowered inspection



Alternative text: monochrome orientation plate showing an inspected joystick module connected to Mega A0, A1, D22, 5 V, and ground; TP-X and TP-Y at the two analog signals; and a separately resistor-limited common-cathode RGB LED on D5, D6, and D7.

Observed role	Reference pin	Connection after inspection
X signal: VRx, X, or HOR	A0 / D54	module signal; TP-X to Mega GND
Y signal: VRy, Y, or VER	A1 / D55	module signal; TP-Y to Mega GND
select: SW, KEY, or SEL	D22	switch-to-ground, internal pull-up
supply and common	5 V and GND	only after rating and continuity checks
RGB preview	D5, D6, D7	each pin through 330 Ω to LED channel

Unpowered boundary. The exact module labels, contact order, rated supply, ground continuity, select topology, three RGB resistors, and LED common polarity must already be resolved before USB power is connected. TP-X and TP-Y must be isolated from the rail and adjacent rows. This fixture includes no motor, relay, external supply, or unidentified module.

3 Calibration contract

Each axis stores five facts: analog pin, observed minimum, center, observed maximum, and dead-zone half-width. Valid calibration requires

$$m < c < M, \quad c - d > m, \quad c + d < M.$$

Here m is the observed minimum, c the center, M the observed maximum, and d the dead zone. These are configured observations, not claims about universal joystick travel.

$$p(r) = \begin{cases} -1000 \frac{c-d-r}{c-d-m}, & r < c-d, \\ 0, & c-d \leq r \leq c+d, \\ 1000 \frac{r-(c+d)}{M-(c+d)}, & r > c+d. \end{cases}$$

The implementation clamps before narrowing, uses 32-bit intermediates, and applies inversion last. A raw value at or beyond an observed endpoint sets **saturated**; it does not assert an electrical fault. Raw values remain in the snapshot even when their normalized position is zero.

3.1 Boundary examples

Use $m = 80$, $c = 520$, $M = 960$, and $d = 40$.

Raw count	Region	Position	Saturated?
80	negative endpoint	-1000	yes
479	negative travel	-2	no
480	dead zone	0	no
560	dead zone	0	no
561	positive travel	2	no
960	positive endpoint	1000	yes

4 Lifecycle, ownership, and sample order

Construction performs no hardware operation. Initialization validates both axis configurations, three distinct pins, and Mega capabilities before claiming X, then Y, then select. Any failure releases the claims from that attempt in reverse order. Repeated successful initialization is inert.

Every `update(now)` reads X once, Y once, then delegates exactly one select sample to the existing `Button`. Its snapshot contains raw and calibrated axes, raw and debounced selection levels, press and release events, and one complete status. Events remain stable until the next update.

```
const adk::Status status = joystick.update (now);
const adk::AnalogJoystickSnapshot view =
    joystick.snapshot ();

showJoystickPosition (view);
showSelectionEvent    (view);
```

Shutdown stops select, Y, then X. It preserves the last raw and interpreted evidence. The retained snapshot reports a not-initialized status. The three input endpoints return to `INPUT`; an RGB presenter is a separate owner and must also be shut down by the sketch.

5 Circuit-native presentation

The RGB LED is not part of `AnalogJoystick`. The narrative sketch maps the stable snapshot to an independent presentation:

- a short blue pulse after all resources initialize proves acquisition;
- red intensity represents negative X, green intensity positive X;

- centered X uses blue at the same Y-derived brightness;
- Y scales the chosen direction's brightness;
- a debounced select event produces a brief all-channel marker.

The same color never carries the only distinction: TP-X and TP-Y retain direct voltage evidence, and the lesson names the selected channel and magnitude in text. Serial raw counts are optional supporting evidence.

5.1 Current bound

One 330 Ω resistor is required per RGB channel. The sketch commands at most one steady direction channel; the short event marker may command three. A conservative zero-forward-voltage bound is

$$I_{\text{channel}} \leq \frac{5.0 \text{ V}}{330 \Omega} = 15.2 \text{ mA}, \quad I_{\text{RGB,total}} \leq 45.5 \text{ mA}.$$

This is a conservative design bound, not a measurement or permission to omit the three resistors. TP-X and TP-Y remain within the inspected module's rated rail; the select signal is high when released and low when pressed for the documented switch-to-ground fixture.

6 Narrative Mega example

```
void loop ()
{
    const adk::TimePoint now (millis ());

    observeJoystick (now);
    decidePreview   ();
    showPreview     (now);
}
```

Objects appear in dependency order: Mega runtime, resource registry, joystick, RGB endpoints, then presenter state. Setup reads as acquire, configure, start. The loop reads as observe, decide, actuate. It takes one joystick snapshot and never reads a pin directly.

7 Predict, observe, interpret

1. With power absent, predict the center-region TP-X and TP-Y voltages and the select levels against the exact inspected pin labels.
2. Apply USB power. Observe the blue acquisition pulse before interpreting any direction preview.
3. Hold the stick at rest. Compare raw counts with TP-X/TP-Y. Derive the center and dead-zone values from repeated observations, not one sample.
4. Move X toward each mechanical limit without force. Compare raw count, normalized sign, direction channel, and TP-X.
5. Repeat for Y and explain how Y changes magnitude without changing X direction.
6. Bounce and hold select. Compare raw level, debounced level, and the single press/release events.
7. Invoke shutdown. Observe RGB inactivity separately from all three input pins returning to their documented modes; then remove USB.

8 Diagnosis and stop conditions

Observation	Stop, inspect, and interpret
label or pin order differs	remove USB; follow the exact board, not this drawing
TP-X or TP-Y outside rails	remove USB; inspect supply, ground, and signal role
axes exchanged	correct role-to-pin configuration; do not rename raw evidence
rest drifts outside dead zone	repeat observations; do not hide faults with a huge zone
direction sign is opposite	use explicit inversion after confirming wiring
multiple selection events	inspect debounce trace and supplied timestamps
preview but no blue pulse	do not claim acquisition; inspect startup sequence
hot part or unstable USB	remove USB immediately; troubleshoot unpowered

9 Exercises

1. Derive the two denominators for an asymmetric axis and explain why one global slope would bias the center.
2. Identify every invalid configuration that can be rejected before a hardware read or claim.
3. Draw the reverse rollback order when Y acquisition fails, then when select acquisition fails.
4. Explain why **saturated** preserves evidence but cannot diagnose a short circuit.
5. Treat the authorized tilt-ball family as an ordinary **DigitalInput** exercise. State why it does not justify a new semantic component.

10 The two-minute joystick adventure

The visible story depends on each prior discovery: move X left to unlock the red stage, move X right to unlock the green stage, then press Select to reach the steady white finish. Y still controls brightness, so the earlier axis lesson remains visible inside every later stage. At 120 seconds the sketch shuts down regardless of position, button state, or completed stage. The RGB lamp going dark is the learner-facing ending; pin release remains a separate bench claim maintained outside this lesson.

Stop rule. Remove USB power for unknown markings, uncertain pin order, an out-of-rail test point, missing resistor, wrong LED polarity, unexpected behavior, a hot part, unstable power, or disagreement between the schematic and breadboard. Do not move wiring while powered.

11 Software evidence

Repository automation verifies deterministic mapping, lifecycle behavior, Mega compilation, size budgets, and publication. Those maintainer gates are supporting evidence, not learner steps and not a physical acceptance result.

12 Reference trail

- ADK `analog_joystick.h`, `analog_input.h`, `button.h`, focused host tests, and canonical Lesson 031 sketch;

- ADK development, testing, safety, packaging, and PDF policies;
- the authorized Elegoo-set manifest for family provenance only;
- the exact module's markings and primary electrical source before powered work;
- Arduino Mega 2560 Rev3 datasheet for the board profile.

The searchable HTML reference is the primary accessible route. This printable PDF uses real text, headings, lists, and monochrome graphics but does not claim PDF/UA conformance.